

MS SQL Server

Introduction

- MS SQL Server is a database server
- Product of Microsoft
- Enables user to write queries and other SQL statements and execute them
- Consists of several features. A few are:
 - Query Analyzer
 - Profiler
 - Service Manager
 - Bulk Copy Program (BCP)

Profiler

- Monitoring tool
- Used for performance tuning
- Uses traces – an event monitoring protocol
- Event may be a query or a transaction like logins etc

Service Manager

- Helps us to manage services
- More than one instance of SQL server can be installed in a machine
- First Instance is called as default instance
- Rest of the instances (16 max) are called as named instances
- Service manager helps in starting or stopping the instances individually

Instances

- Each instance is hidden from another instance
- Enhances security
- Every instance has its own set of Users, Admins, Databases, Collations
- Advantage of having multiple instance is
 - Multi company support (Each company can have its own instance and create databases on the same server, independent on each other)
 - Server consolidation (Can host up to 10 server applications on a single machine)

BCP

- Bulk Copy Program
- A powerful command line utility that enables us to transfer large number of records from a file to database
- Time taken for copying to and from database is very less
- Helps in back up and restoration

Query Analyzer

- Allows us to write queries and SQL statements
- Checks syntax of the SQL statement written
- Executes the statements
- Store and reload statements
- Save the results in file
- View reports (either as grid or as a text)

SQL Database Objects

- A SQL Server database has lot of objects like
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - Rules
 - Defaults
 - Cursors
 - Triggers

System Databases

- By default SQL server has 4 databases
 - Master : System defined stored procedures, login details, configuration settings etc
 - Model : Template for creating a database
 - Tempdb : Stores temporary tables. This db is created when the server starts and dropped when the server shuts down
 - Msdb : Has tables that have details with respect to alerts, jobs. Deals with SQL Server Agent Service

Creating a database

- We need to use Master database for creating a database
- By default the size of a database is 1 MB
- A database consists of
 - Master Data File (.mdf)
 - Primary Log File (.ldf)

Database operations

- Changing a database
Use <dbname>
- Creating a database
Create database <dbname>
- Dropping a database
Drop database <dbname>

SQL Server Data types

- Integer : Stores whole number
- Float : Stores real numbers
- Text: Stores characters
- Decimal: Stores real numbers
- Money : Stores monetary data. Supports 4 places after decimal
- Date : Stores date and time
- Binary : Stores images and other large objects
- Miscellaneous : Different types special to SQL Server.
(Refer to notes for more info)

Operators

- Arithmetic
- Assignment
- Comparison
- Logical
- String
- Unary
- Bitwise

Select Statements

- **To execute a statement in MS SQL, Select the statement and Click on the Execute button in the query analyser or press F5**
- This is used to retrieve records from a table
- Eg. Select * from table1;
 - This will fetch all rows and all columns from table1
- Eg. Select col1,col2 from table1
 - This will fetch col1 and col2 from table1 for all rows
- Eg. Select * from table1 where <<condn>>
 - This will fetch all rows from table1 that satisfies a condition
- Eg. Select col1,col2 from table1 where <<condn>>
 - This will fetch col1 and col2 of rows from table1 that satisfies a condition

Select Options

- Aggregate functions
 - Sum(col1): sum of data in the column col1
 - Max(col1): data with maximum value in col1
 - Min(col1): data with minimum value in col1
 - Avg(col1): Average of data in col1
 - Count(col1): Number of not null records in table
- Grouping – Group by col1 : Groups data by col1
- Ordering – Order by col1 : Orders the result in ascending order (default order) of col1
- Filtering – Where <<condn>> and Having <<condn>>

Table management

Create table tablename

```
(  
    col1    data type,  
    col2    data type  
);
```

- Creates a table with two columns

Drop table tablename;

- Drops the table structure

Insert statements

- Inserting data to all columns
 - Insert into tablename(col1,col2) values(v1,v2)
 - Insert into tablename values(v1,v2)
- Inserting data to selected columns
 - Insert into tablename(col1) values (v1)
 - Insert into tablename(col2) values (v2)

Update statement

Update table tablename

Set colname=value

- This updates all rows with colname set to value

Update table tablename

Set colname=value

Where <<condition>>

- This updates selected rows with colname as value only if the row satisfies the condition

Delete statements

Delete from table1;

Deletes all rows in table1

Delete from table1 where <<condition>>

Deletes few rows from table1 if they satisfy the condition

Truncate statement

- Truncate table tablename
- Removes all rows in a table
- Resets the table.
- Truncate does the following, where as delete statement does not
 - Releases the memory used
 - Resets the identity value
 - Does not invoke delete trigger

Alter statements

- Used to modify table structure
 - Add new column
 - Change data type of existing column
 - Delete a column
 - Add or remove constraints like foreign key, primary key

Table name

Attribute names

Tables in SQL

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Tuples or rows